COPO

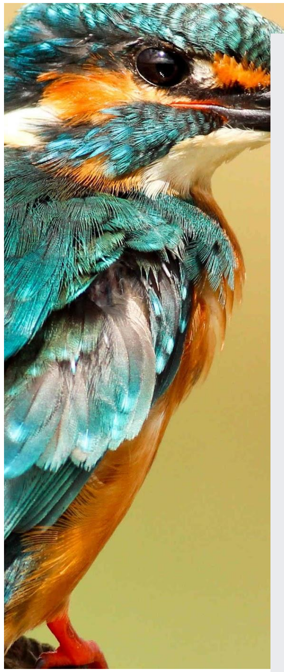
VISUAL DOCUMENTATION FOR MANIFEST SUBMISSION

Collaborative OPen Omics

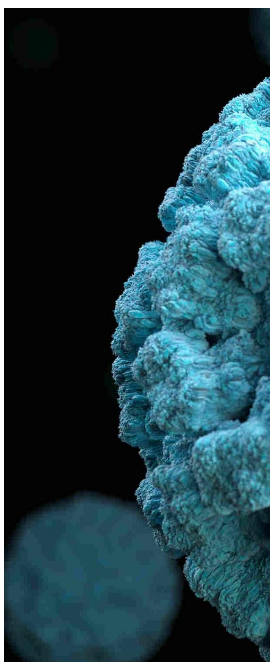
ei.copo@earlham.ac.uk, @copo-project

Earlham Institute

Decoding Living Systems




Contents **Slide #**




Prerequisites:

- An ORCid account
- Contact ei.copo@earlham.ac.uk to be added to a project manifest group
- Ensure that you have the latest version of the manifest if you are submitting samples as part of -
  - ➔ Aquatic Symbiosis Genomics (ASG) project, please refer to the latest metadata manifest
  - ➔ Darwin Tree of Life (DToL) project, please refer to the latest metadata manifest here
  - ➔ European Reference Genome Atlas (ERGA) project, please refer to the latest metadata manifest here

ℹ The latest manifest templates can also be found in our documentation.

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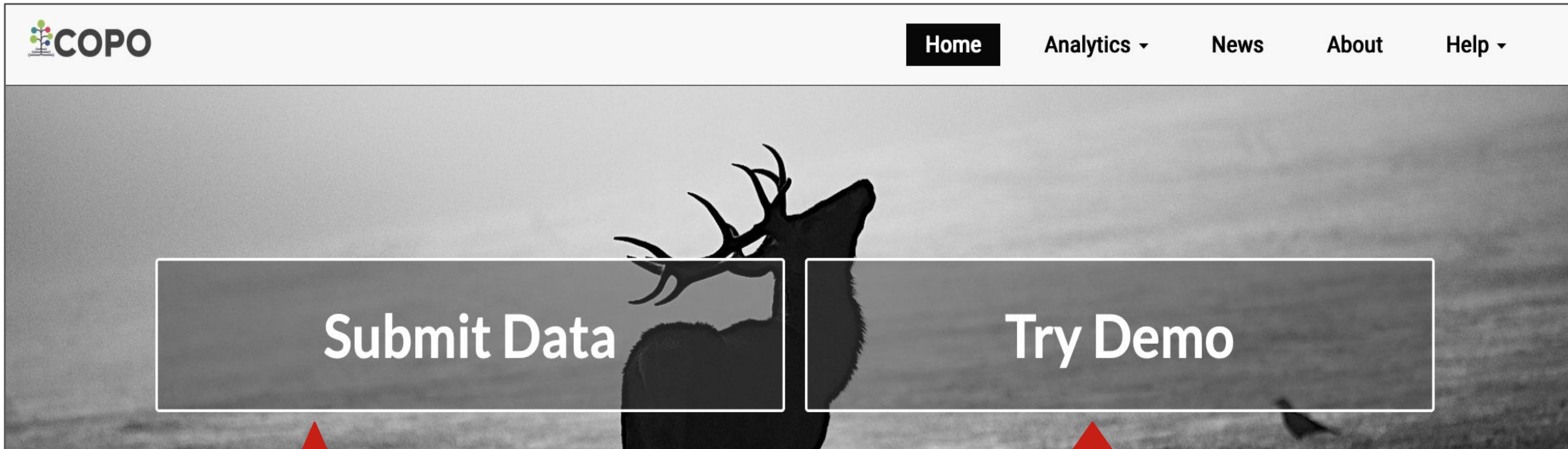
COPO’s home page:



To familiarise yourself with the platform

IMPORTANT - Submissions to European Nucleotide Archive (ENA) through the demo server will not persist, accessions created will be deleted after 24 hours.

The demo server is frequently updated so it may occasionally be unreachable and you may find that we have deleted your past uploads.

Once you are confident

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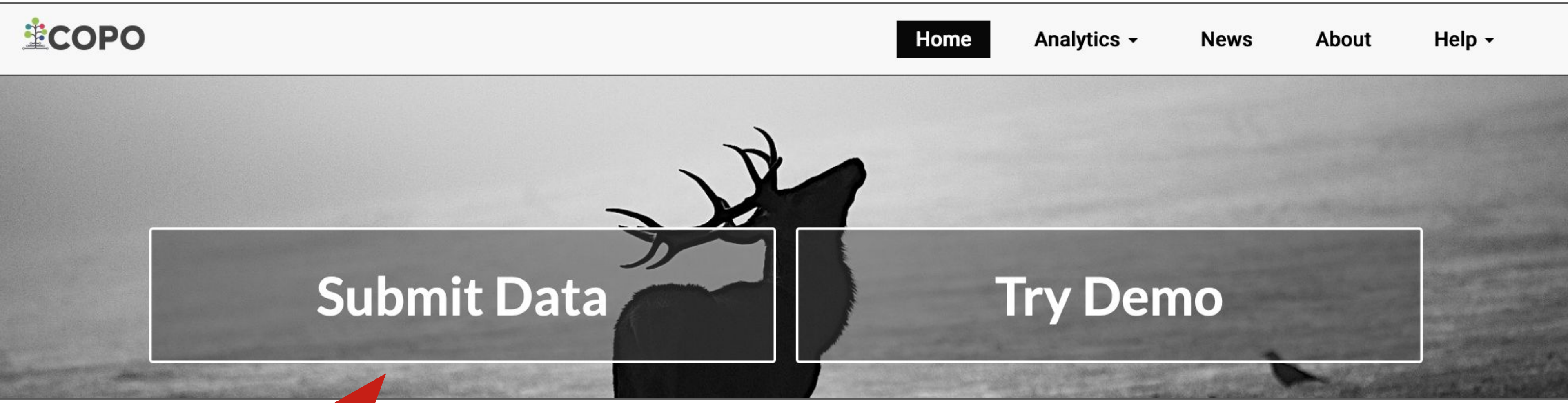
Click “Submit Data” button to log in with your ORCiD credentials:



An ORCiD sign-in form will be displayed after you click the **Submit Data** button.

Enter your ORCiD credentials to proceed to the COPO **Work profiles** web page.

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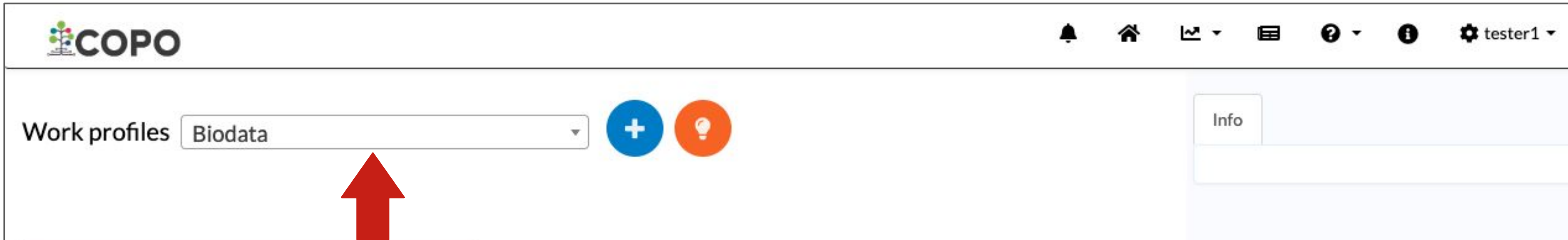
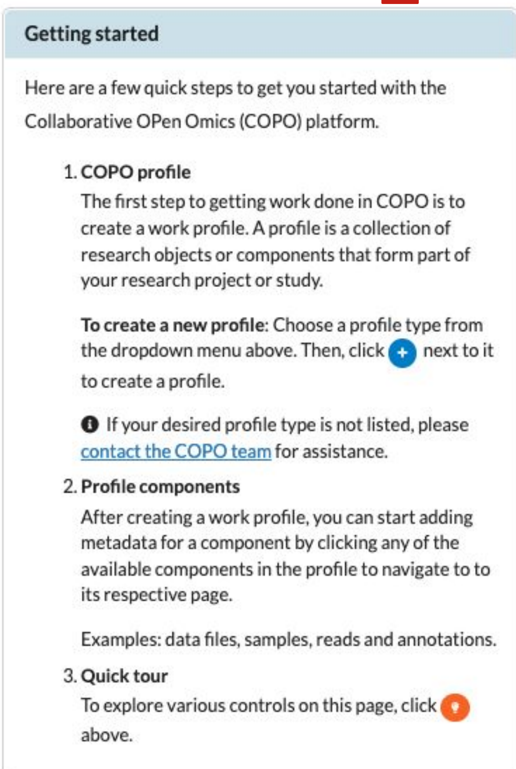
Choose profile type from dropdown menu:





IMPORTANT

- If you cannot see the project that you are associated with in the dropdown menu, you need to be added to the relevant group.

  Please stop here and get in touch with ei.copo@earlham.ac.uk.

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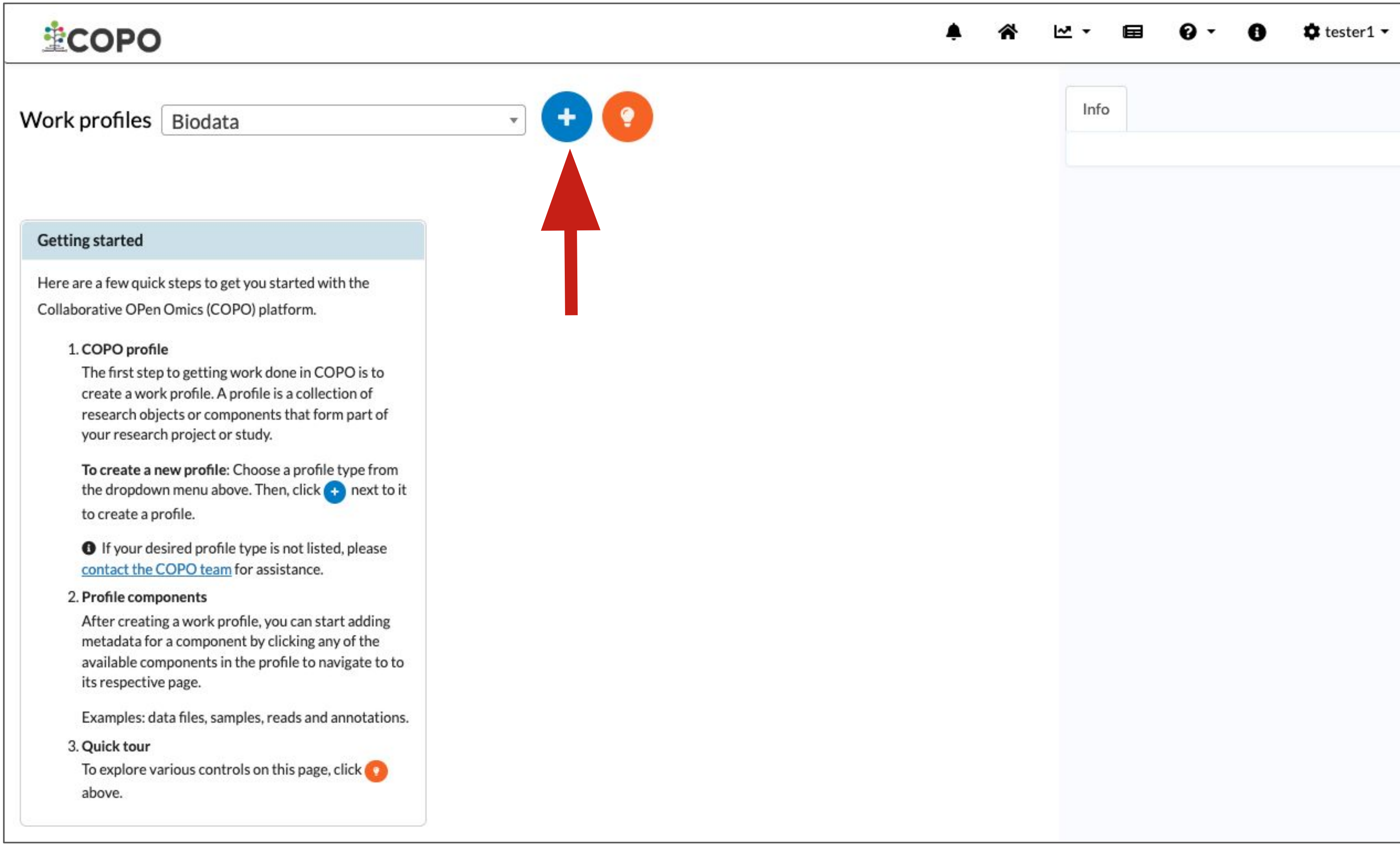
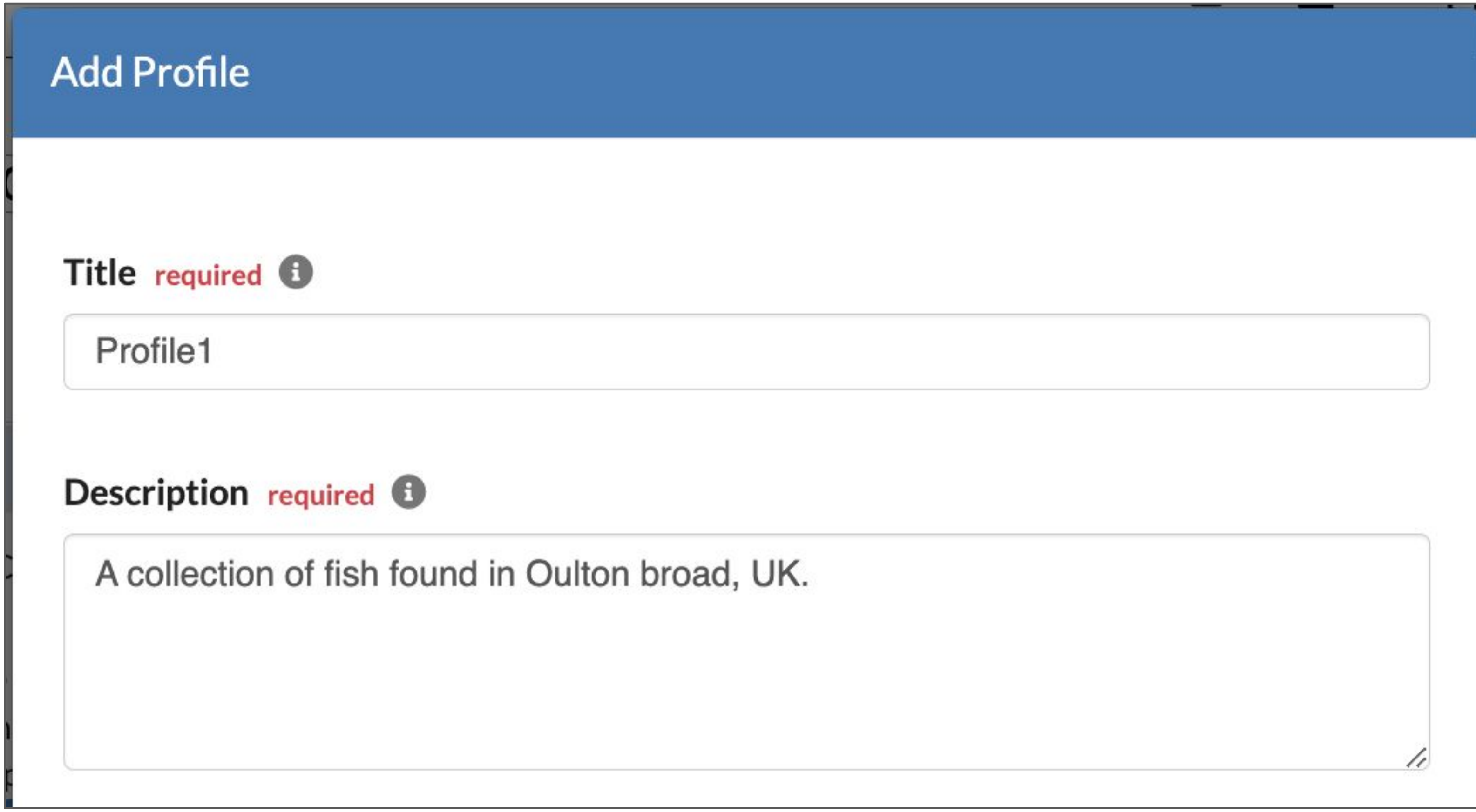
Create a profile:



An **Add Profile** form will be displayed after you click the blue button that has a white plus sign in its centre.

Create a profile:



IMPORTANT

- Name the profile based on the sample set identifier (ID) assigned to the samples that you are submitting.

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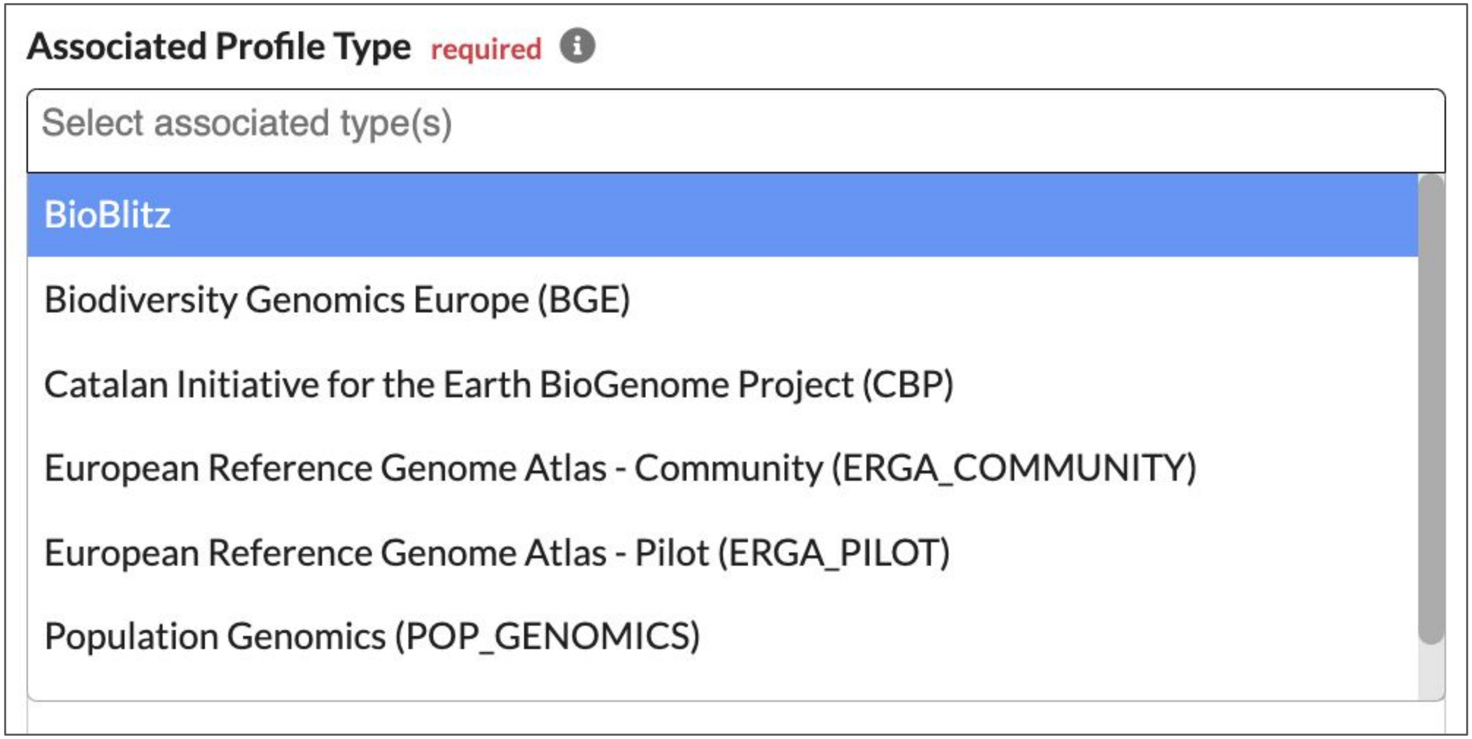
Add associated profile type (if applicable)



NOTE - for ERGA profile types only

- If your desired profile type is a part of a larger project, select the associated type from the **Associated Profile Type** dropdown menu.

- More than one associated type can be selected from the dropdown menu for the profile type.

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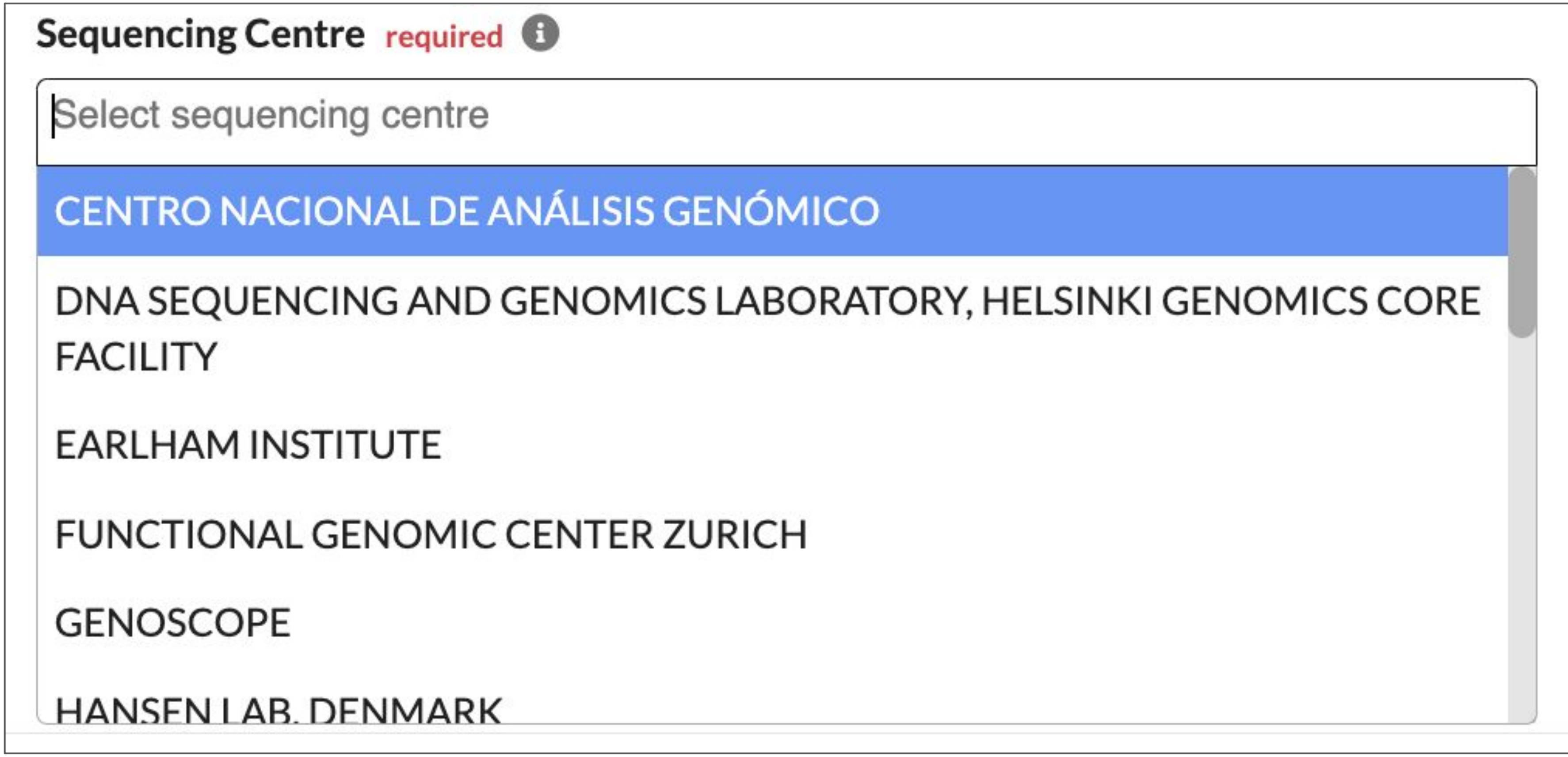
Add sequencing centre (if applicable):



NOTE - for ERGA profile types only

- Select the the centre where your samples will be sequenced at from the **Sequencing Centre** dropdown menu.

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Add locus tags (if applicable):

More information about locus tags can be found in the FAQ of the COPO documentation

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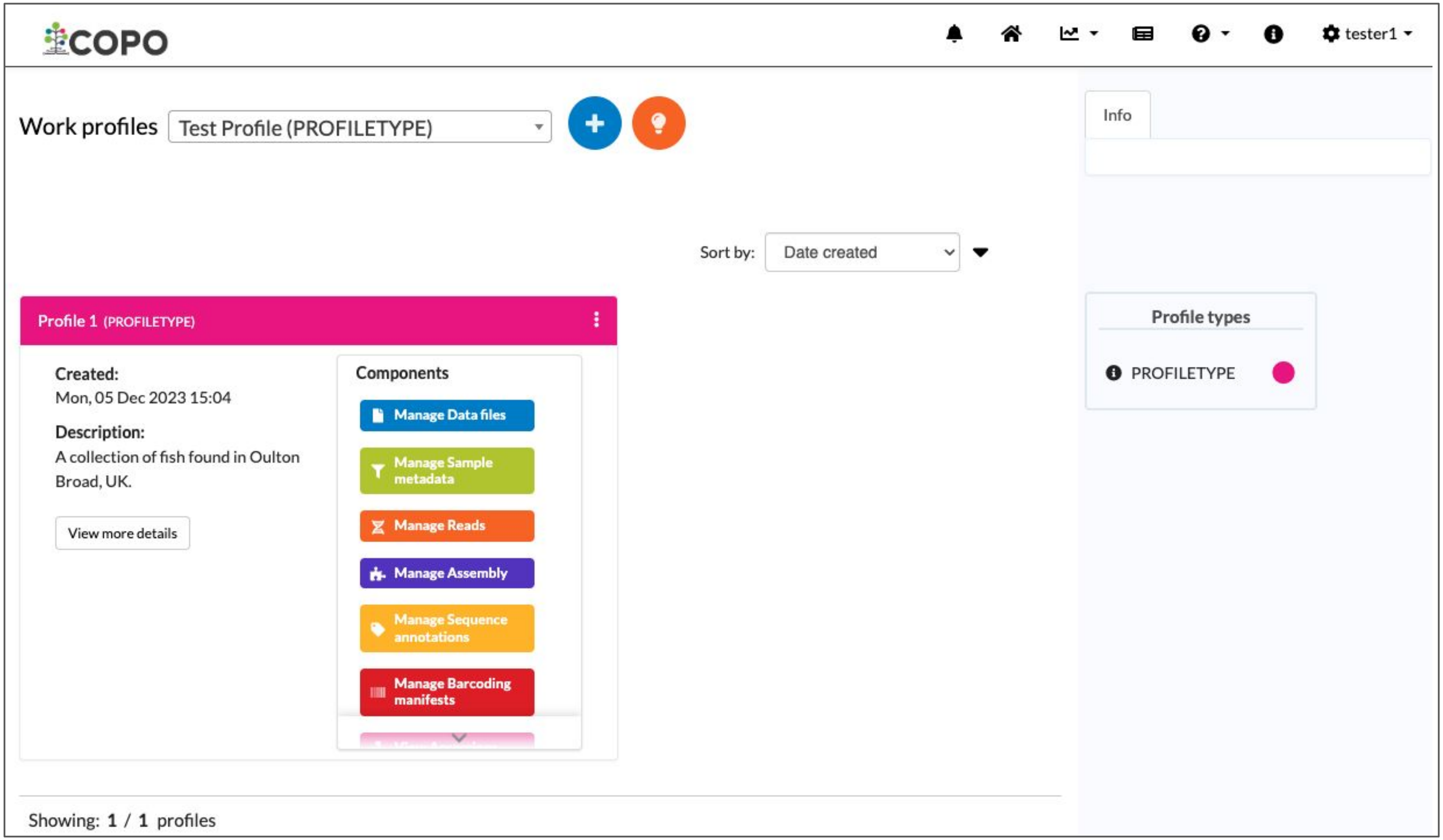
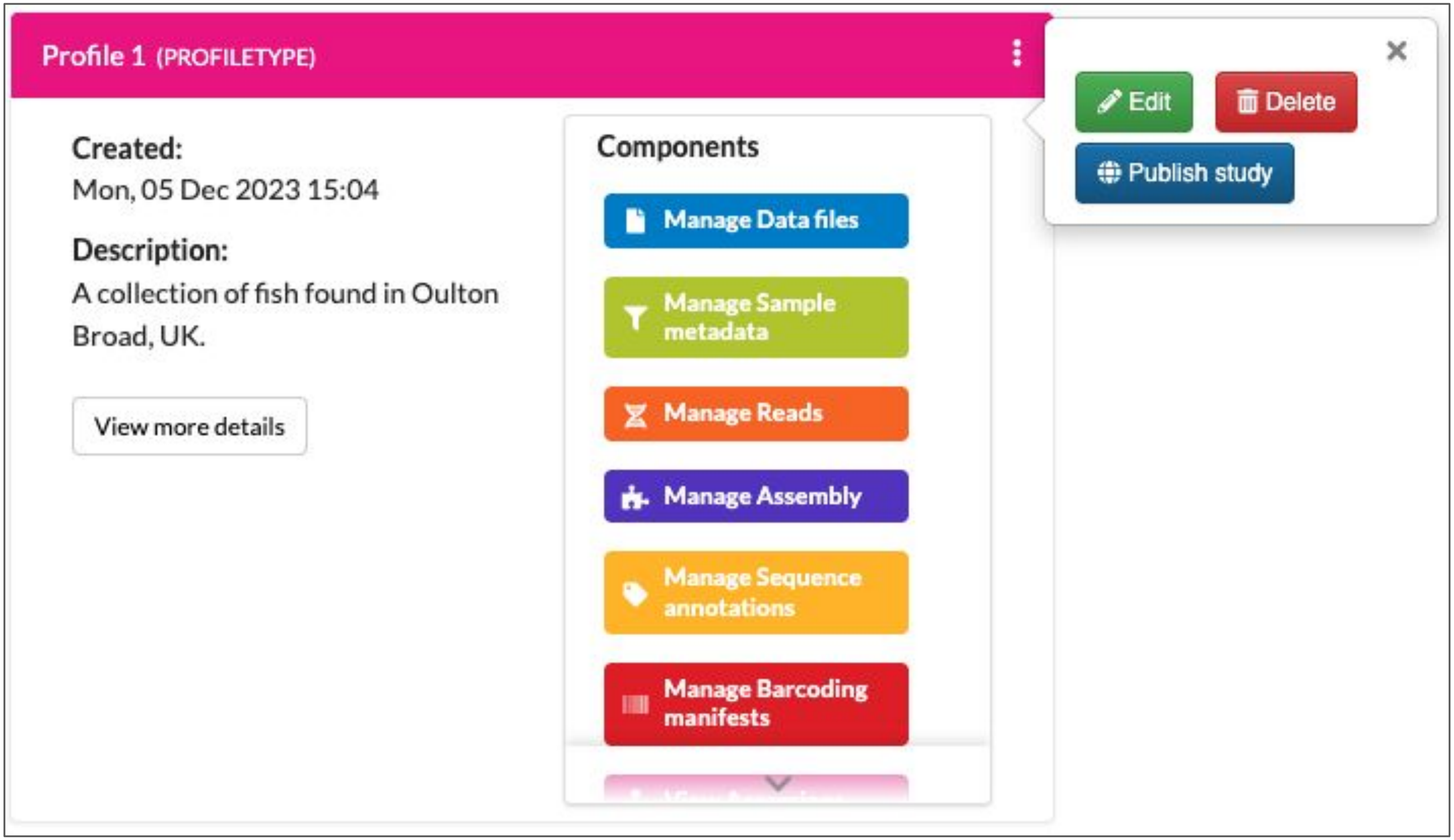
New profile created:



After having provided the details in the **Add Profile** form and clicked the **Save** button, your new profile will be displayed on the COPO **Work profiles** web page

Edit or Delete profile:



If you would like to edit or delete a profile that you have created, click the white vertical ellipsis (⋮) located at the top right of the desired profile panel to view the profile options to do so

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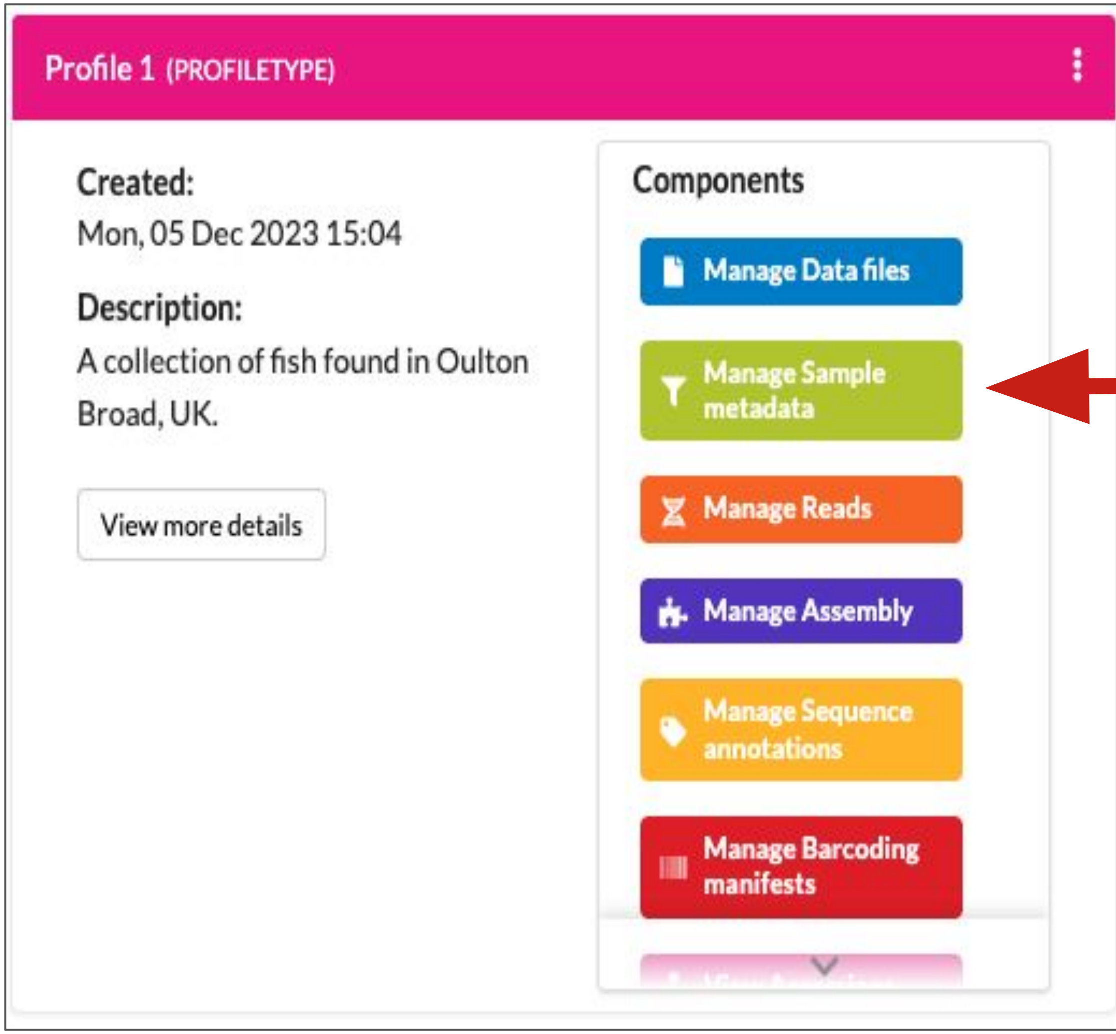
Upload a manifest:



The colour and wording of this profile panel will differ depending on which project you are submitting a manifest for.

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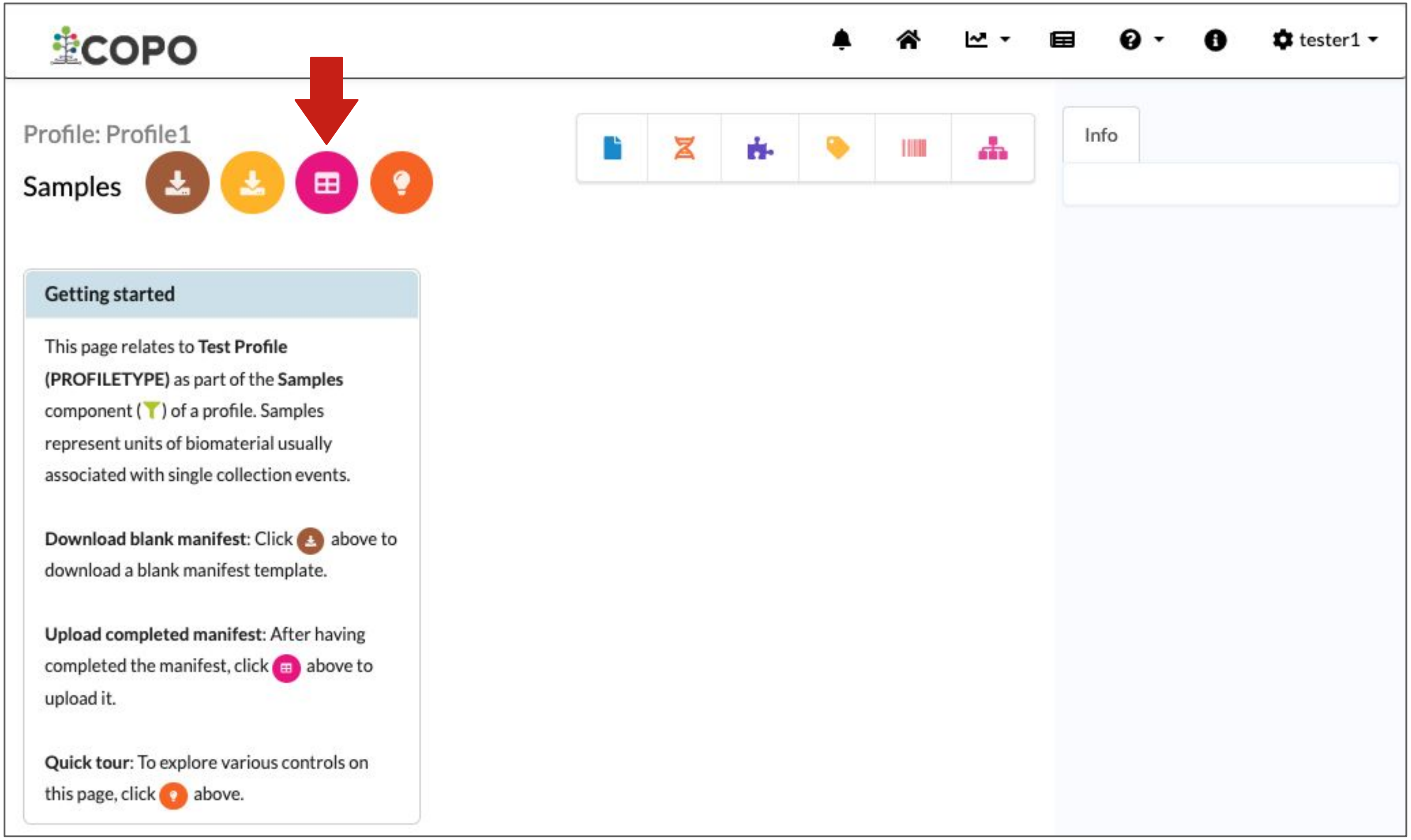
Upload a manifest:



The colour and wording of the **Samples** web page will differ depending on which project you are submitting a manifest for.

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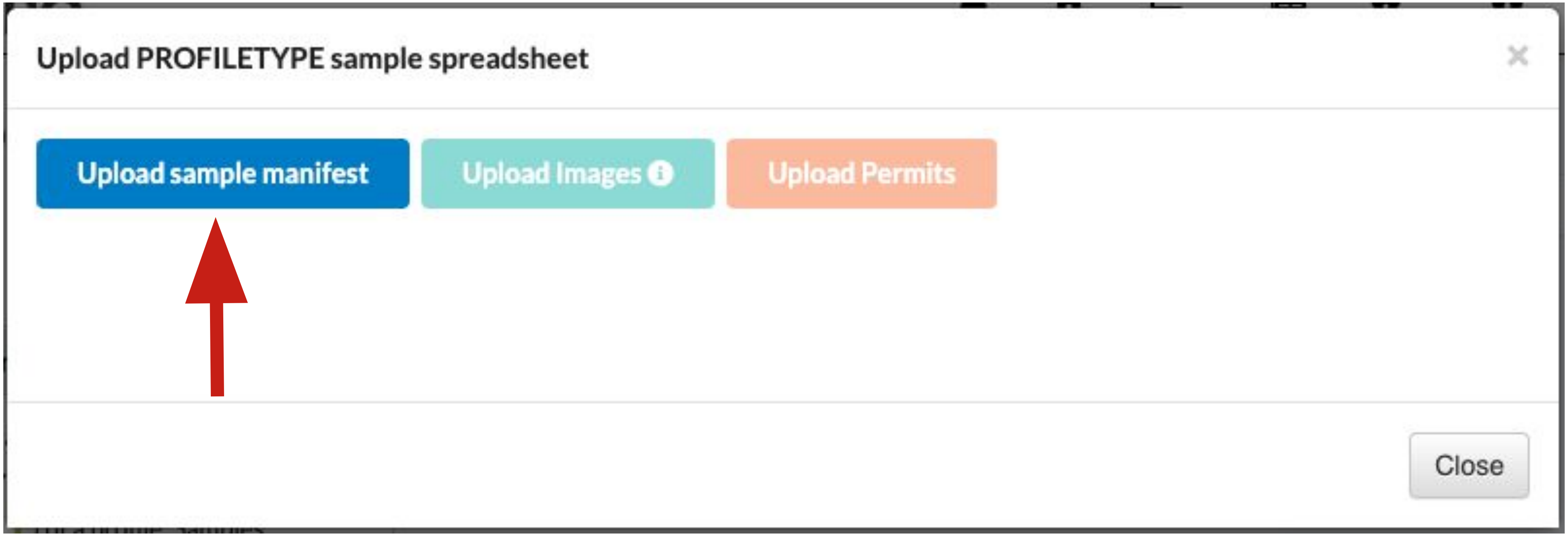
Upload a manifest:



COPO will validate taxonomy information and manifest format:

What's happening (part 1):

- COPO is retrieving information from National Centre for Biotechnology Information (NCBI) to check that the taxonomy information provided is correct. We also check that the TAXON_ID is submittable to ENA.
- If some of the taxonomy fields are missing (excluding SCIENTIFIC_NAME), COPO will make an educated guess about what these fields should be and will display warnings about what is going to be filled in.

  The taxonomy checks will be much quicker if you fill in TAXON_ID.
- COPO will substitute scientific name synonyms with the official scientific name. It will display a warning and also record the change in OTHER_INFORMATION.
- If any taxonomy field is incorrect, COPO will display an error and reject the manifest. You can download the error list with the red "Export Errors" button on the top right.
- **If you are unhappy with any of the warnings or errors, DO NOT click the finish button instead, contact NCBI to discuss the taxonomy.**

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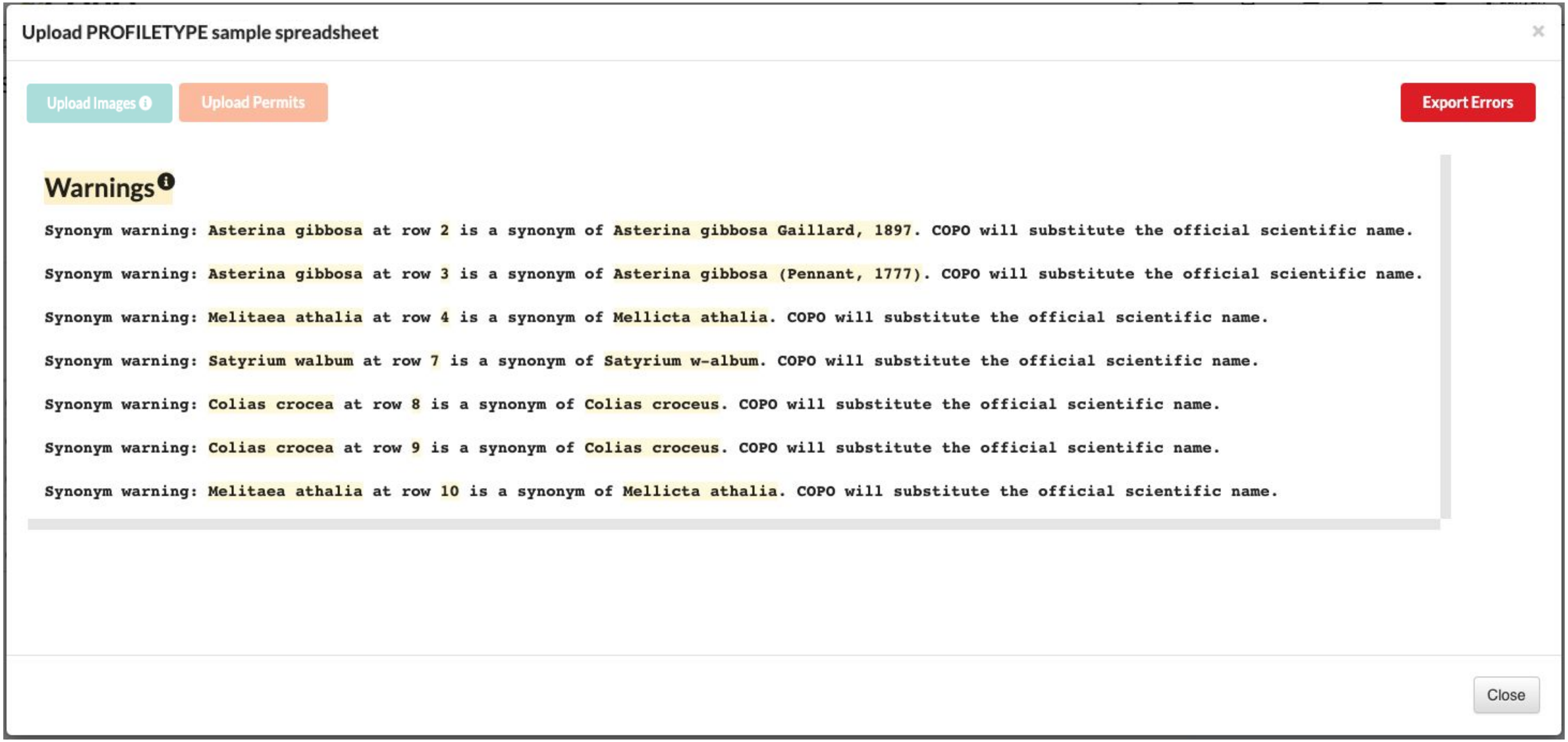
Manifest uploaded -taxonomy warnings-:



These are warnings. COPO will make these changes

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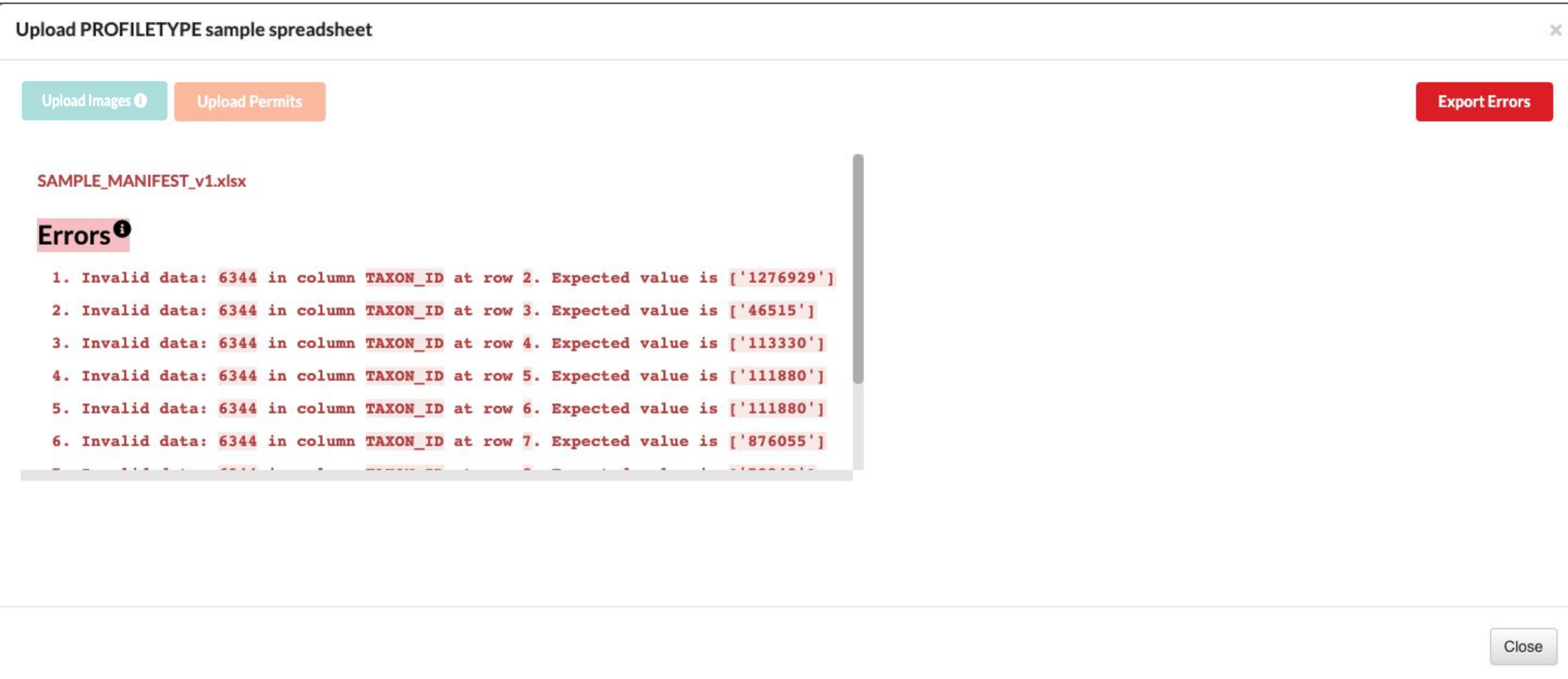
Manifest uploaded -taxonomy errors-:



These are errors. COPO will not accept the manifest

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What’s happening (part 2):

- **ONLY IF** the taxonomy validation was successful and produced no errors, will COPO check the manifest against Standard Operating Procedure (SOP) specifications.

  This includes checks for general format, missing data, cells that have restricted values, etc..

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Manifest uploaded - with compliance errors-:

Upload PROFILETYPE sample spreadsheet

Upload Images | Upload Permits | Export Errors

This manifest was compliant with an older version of the SOP and as such cannot be accepted

SAMPLE_MANIFEST_v2.xlsx

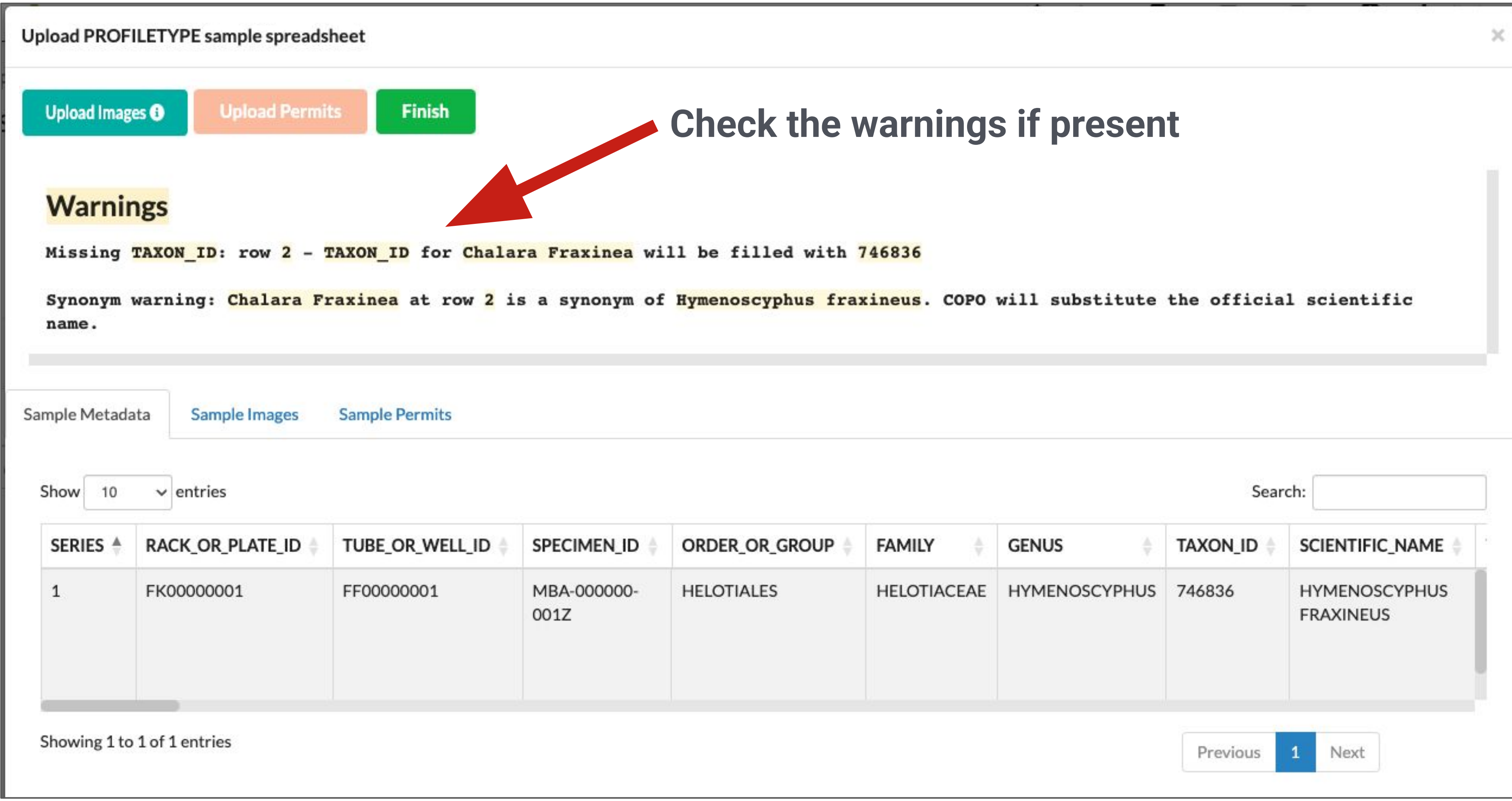
Errors

1. Field not found - SPECIMEN_IDENTITY_RISK
2. Invalid data: REFERENCE GENOME in column PURPOSE_OF_SPECIMEN at row 2. Allowed values are ['REFERENCE_GENOME', 'RESEQUENCING', 'DNA_BARCODING_ONLY', 'RNA_SEQUENCING', 'R&D', 'NOT_PROVIDED']
3. Invalid data: WHOLE ORGANISM in column ORGANISM_PART at row 2. Allowed values are ['**OTHER_FUNGAL_TISSUE**', '**OTHER_PLANT_TISSUE**', '**OTHER_REPRODUCTIVE_ANIMAL_TISSUE**', '**OTHER_SOMATIC_ANIMAL_TISSUE**', 'ABDOMEN', 'ANTERIOR_BODY', 'BLADE', 'BLOOD', 'BODYWALL', 'BRACT', 'BRAIN', 'BUD', 'CAP', 'CEPHALOTHORAX', 'EGG', 'EGGSHELL', 'ENDOCRINE_TISSUE', 'EYE', 'FAT_BODY', 'FIN', 'FLOWER', 'GILL_ANIMAL', 'GILL_FUNGI', 'GONAD', 'HAIR', 'HEAD', 'HEART', 'HEPATOPANCREAS', 'HOLDFAST_FUNGI', 'INTESTINE', 'KIDNEY', 'LEAF', 'LEG', 'LIVER', 'LUNG', 'MID_BODY', 'MODULAR_COLONY', 'MOLLUSC_FOOT', 'MULTICELLULAR_ORGANISMS_IN_CULTURE', 'MUSCLE', 'MYCELIUM', 'MYCORRHIZA', 'OVARY_ANIMAL', 'OVIDUCT', 'PANCREAS', 'PETIOLE', 'POSTERIOR_BODY', 'ROOT', 'SCALES', 'SCAT', 'SEED', 'SEEDLING', 'SHOOT', 'SKIN', 'SPERM_SEMINAL_FLUID', 'SPLEEN', 'SPORE', 'SPORE_BEARING_STRUCTURE', 'STEM', 'STIPE', 'STOMACH', 'TENTACLE', 'TERMINAL_BODY', 'TESTIS', 'THALLUS_FUNGI', 'THALLUS_PLANT', 'THORAX', 'UNICELLULAR_ORGANISMS_IN_CULTURE', 'WHOLE_ORGANISM', 'WHOLE_PLANT', 'NOT_APPLICABLE', 'NOT_COLLECTED', 'NOT_PROVIDED']

Close

The errors should guide you to correct the manifest. You will have to correct and re-upload the updated spreadsheet

Manifest uploaded -correct manifest-:



The fields displayed in this table will differ depending on which project you are submitting a manifest for.

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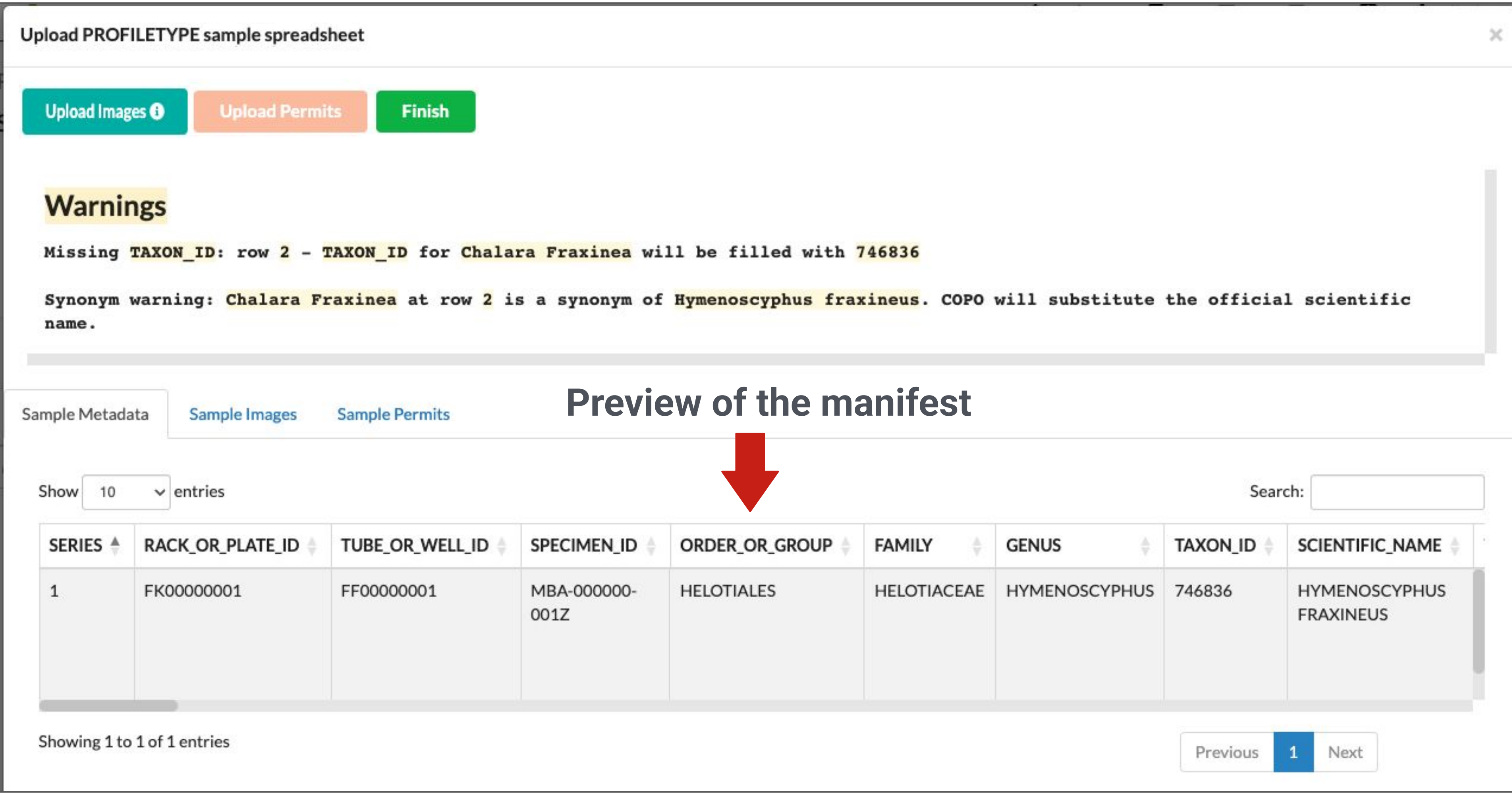
Manifest uploaded -correct manifest-:



SERIES	RACK_OR_PLATE_ID	TUBE_OR_WELL_ID	SPECIMEN_ID	ORDER_OR_GROUP	FAMILY	GENUS	TAXON_ID	SCIENTIFIC_NAME
1	FK00000001	FF00000001	MBA-000000-001Z	HELOTIALES	HELOTIACEAE	HYMENOSCYPHUS	746836	HYMENOSCYPHUS FRAXINEUS

The **Upload Permits** button will only be enabled at this point if permits are required and have been specified in the uploaded manifest.

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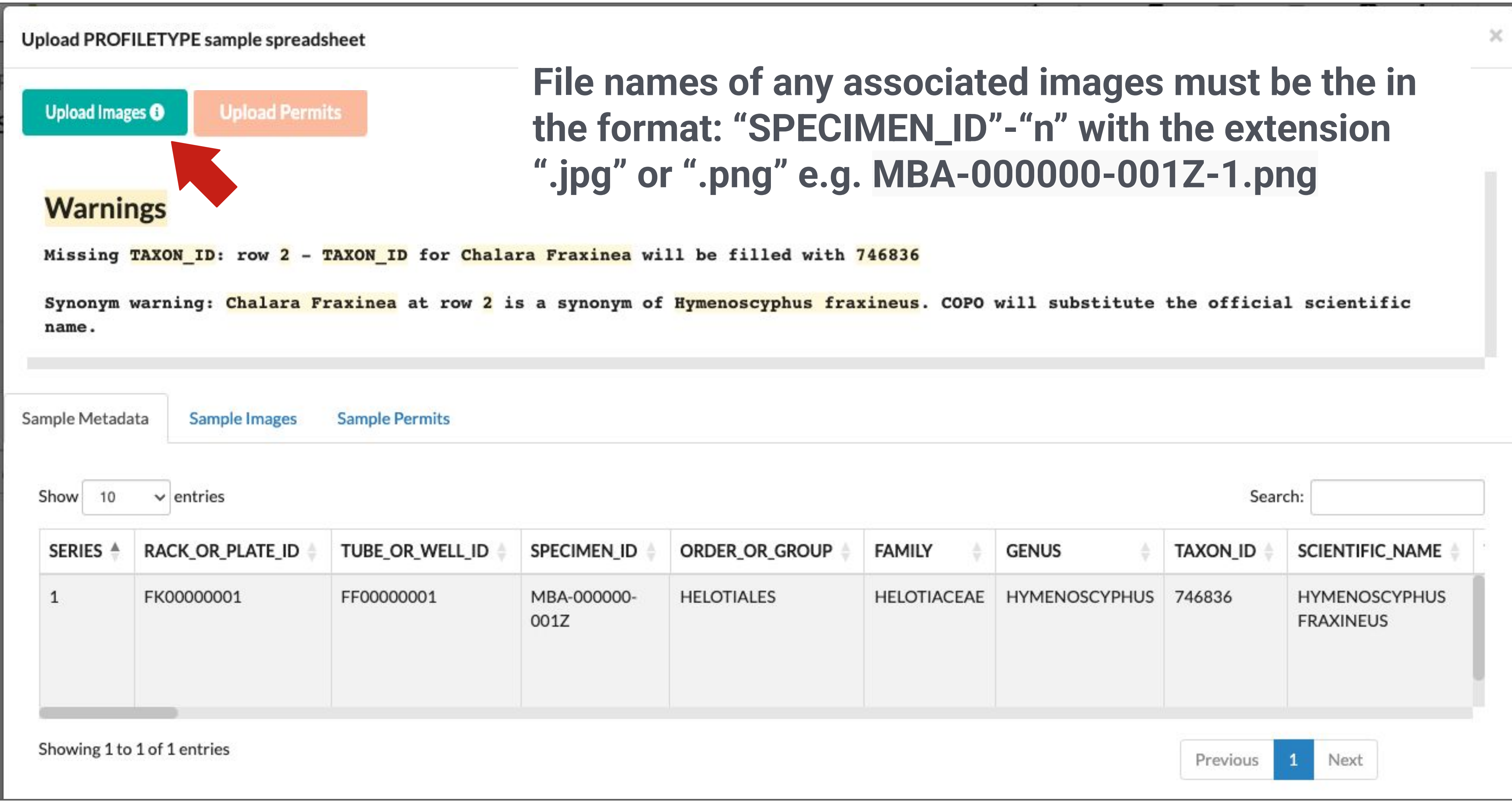
Upload images:

File names of any associated images must be the in the format: “SPECIMEN_ID”-“n” with the extension “.jpg” or “.png” e.g. MBA-000000-001Z-1.png



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Uploaded images:

Upload PROFILETYPE sample spreadsheet ×

Upload Images
Upload Permits

Warnings

Missing TAXON_ID: row 2 - TAXON_ID for Chalara Fraxinea will be filled with 746836

Synonym warning: Chalara Fraxinea at row 2 is a synonym of Hymenoscyphus fraxineus. COPO will substitute the official scientific name.

Sample Metadata | Sample Images | Sample Permits

Show 10 entries

Search:

Specimen ID	Image File	Image
MBA-000000-001Z	MBA-000000-001Z-1.png	COPO

Showing 1 to 1 of 1 entries

Previous 1 Next

Upload permits:

File names (including the extension) of any associated permits must be in the format identified in the "Permit type"_FILENAME column in the spreadsheet e.g. permit1.pdf

IMPORTANT - If you have more than one permit file to upload, they must be uploaded at the same time i.e. after you have clicked the **Upload Permits** button, navigate to the directory where the permits are stored and **CTRL + click** all of the permits so that all the permits are highlighted and uploaded at the same time.

All permit files have to be selected/opened from the directory and uploaded together not one after the other.

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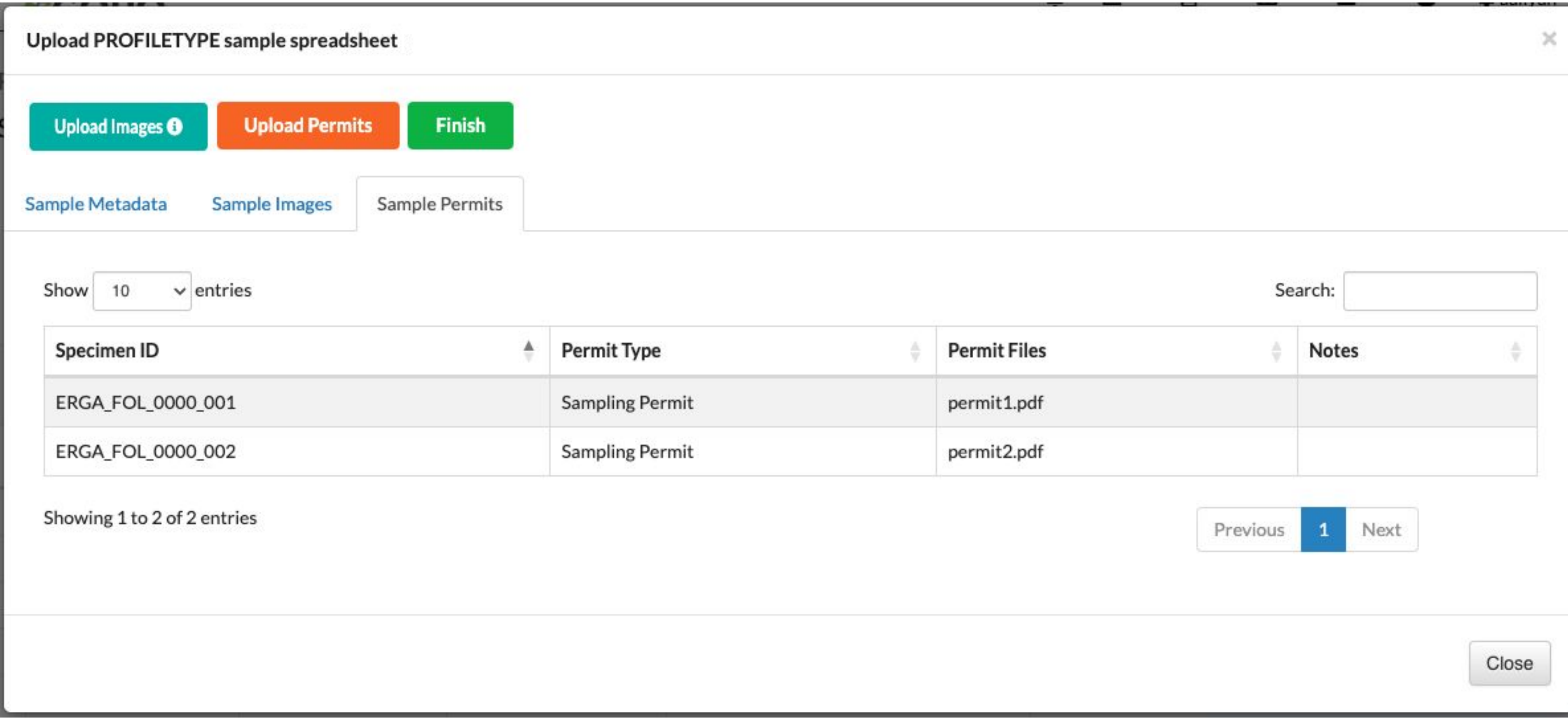
Uploaded permits:



If there are any errors with the permit upload process, they will be displayed in the **Notes** table column under the **Sample Permits** tab

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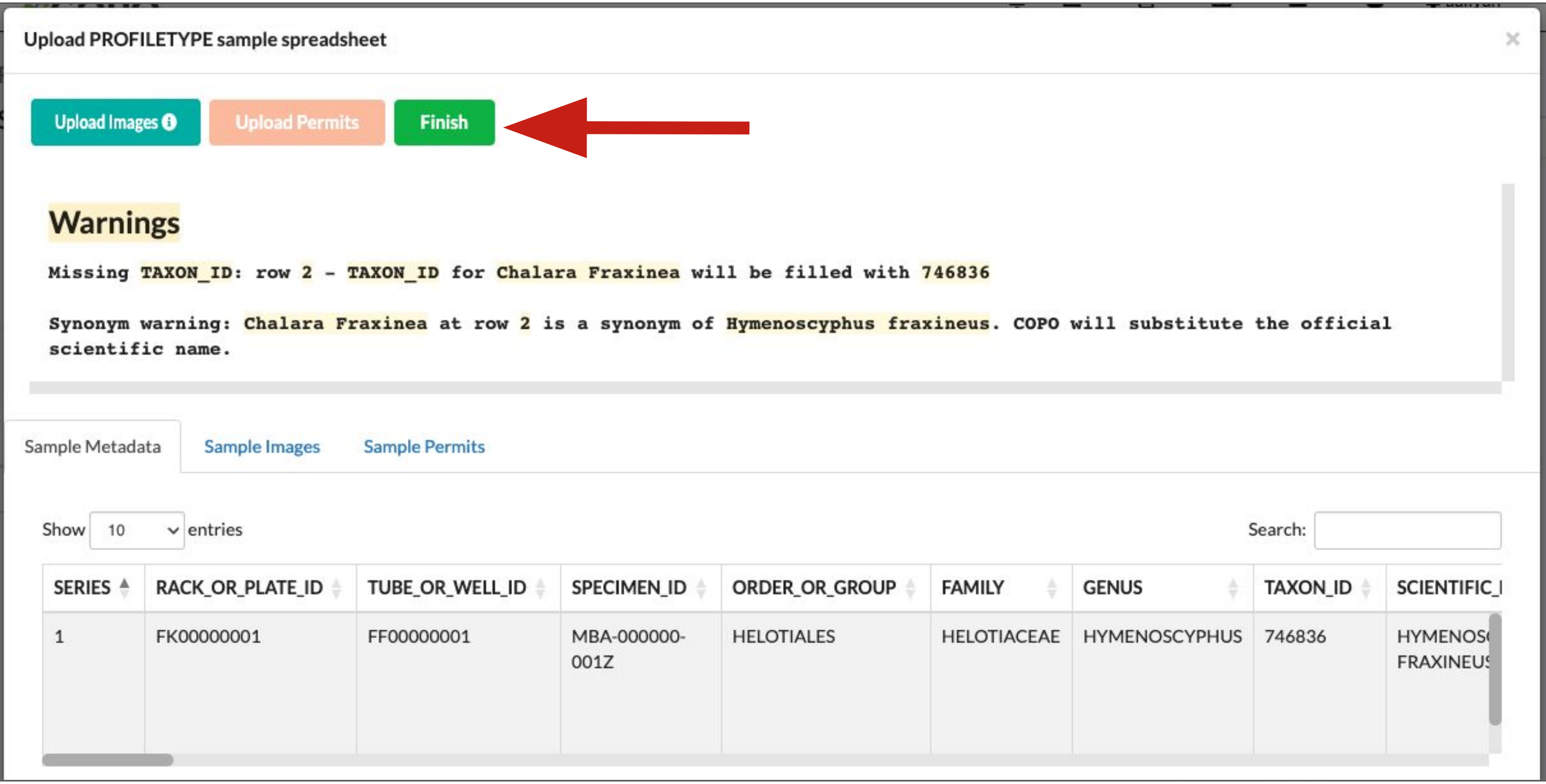
Create the samples in COPO by clicking the “Finish” button:



The **Finish** button will only appear if there are no errors in the manifest and, if required, after permits have been uploaded.

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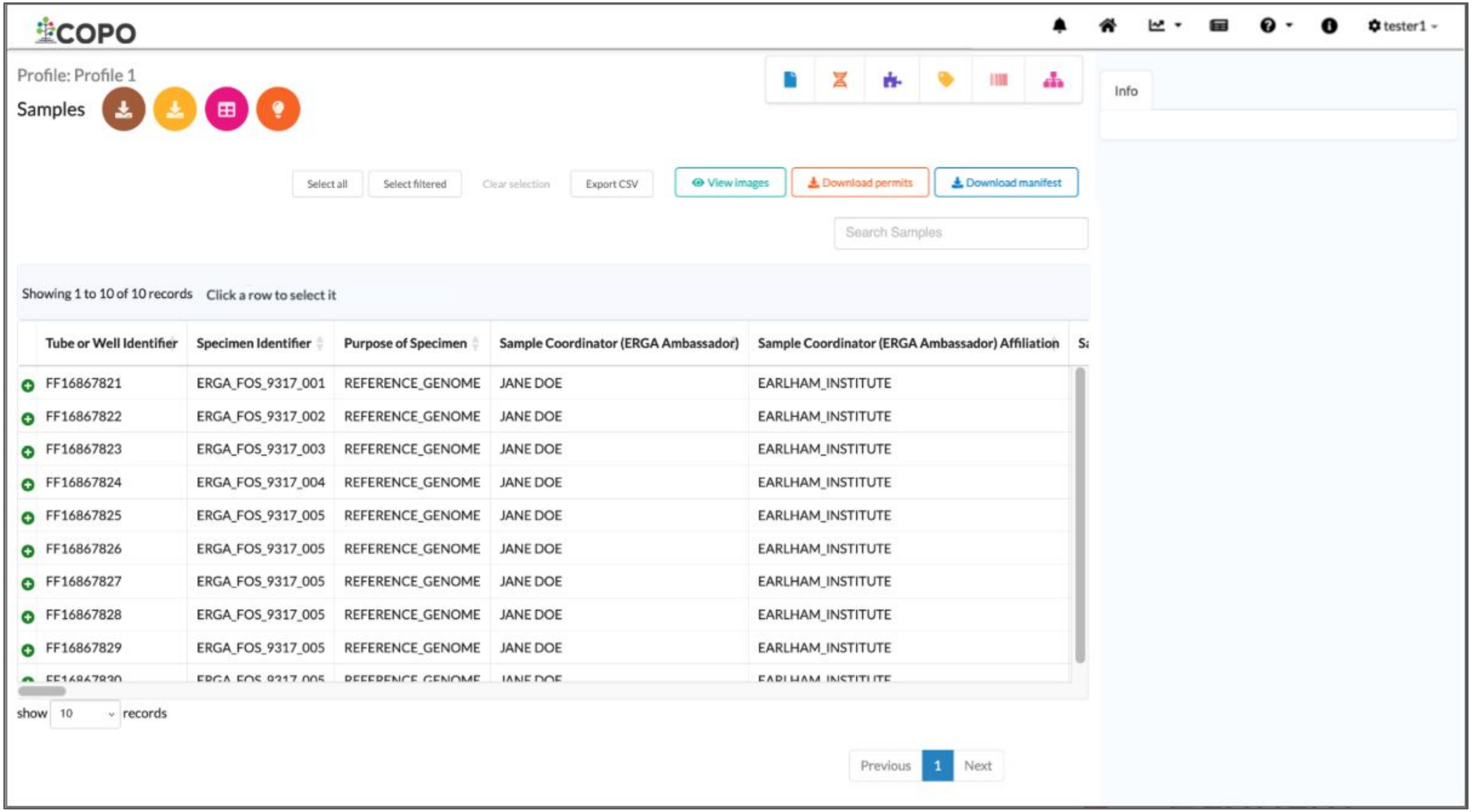
Uploaded samples:



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Sample Managers:

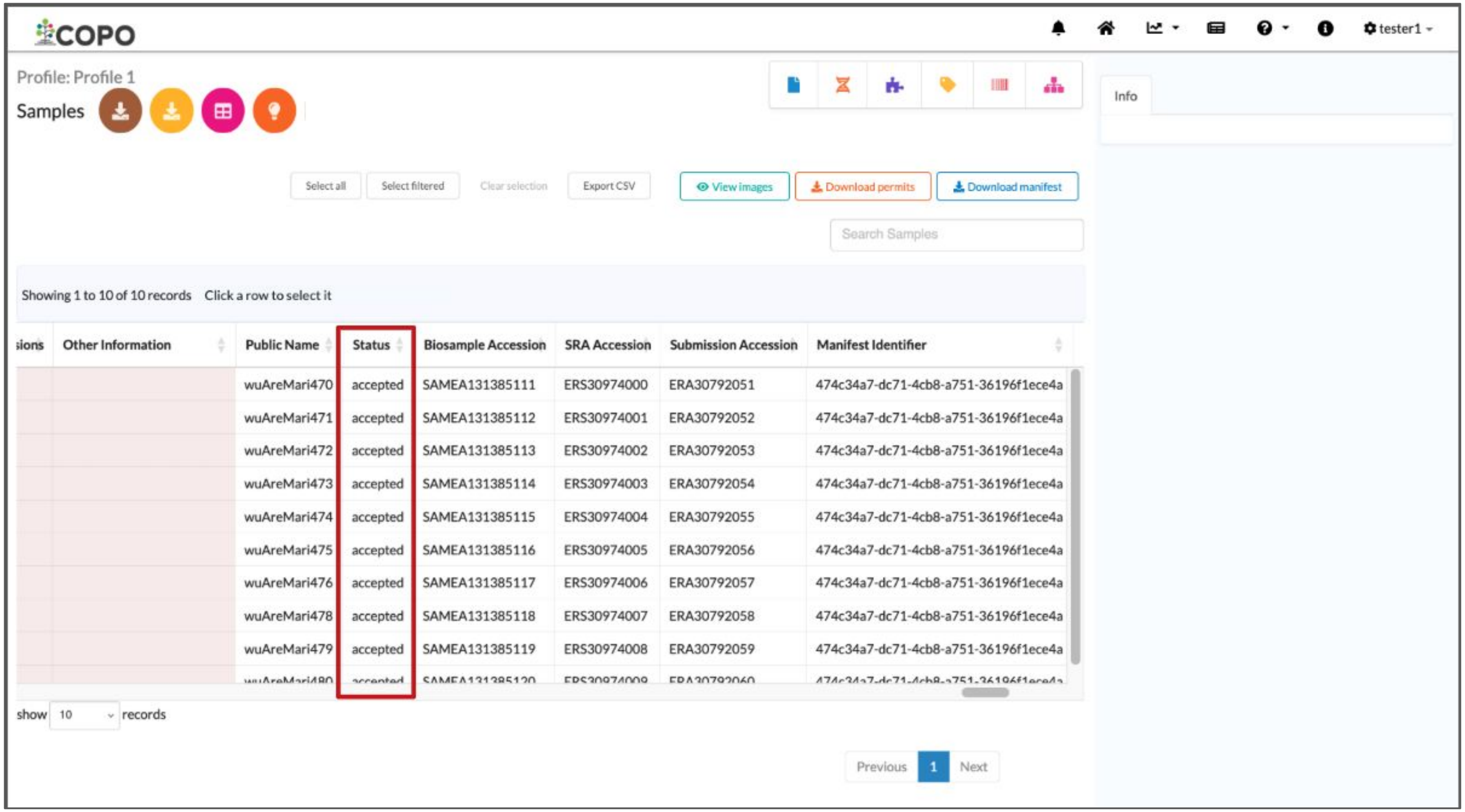
- At this point, the sample managers will be emailed that a new manifest has been uploaded. They will be able to accept or reject each sample after inspection.

  After samples are accepted by a sample manager, they will be submitted to European Nucleotide Archive (ENA).

  Accessions can be inspected via any of the following options:

  - Scrolling right in the previous view
  - Clicking the button on the **Samples** web page to access the **Accessions** web page for the desired profile
  - Clicking the View Accessions component button associated with the desired profile on the **Work profiles** web page to access the **Accessions** web page

Accepted samples:



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Updating samples:

- To update samples, reupload a manifest with the amended changes. The new value will be detected and updated automatically once the resubmission is successfully completed.

  More information regarding sample updates can be found here.

Not all field values of existing samples can be updated by resubmitting a manifest with the amendments.

The fields that can be updated are listed here.

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If you encounter any errors or unexpected behaviour, please email ei.copo@earlham.ac.uk or open a GitHub issue in the COPO repository.

If you have a feature request for future releases, please open an issue in the GitHub repository.

For detailed information regarding manifest submission and COPO, please see our documentation.

EarlhamInst/COPO-production

The Collaborative OPen Omics (COPO) project is supported by the Biotechnology and Biological Sciences Research Council (BBSRC), part of UK Research and Innovation, through the Cellular Genomics (CELLGEN) Grant BBS/E/ER/230001A, Data Infrastructure and Algorithms Group Grant EI-G-Data, Decoding Biodiversity (DECODE): Development of Novel Experimental and Bioinformatic Tools for Genomic Diversity and Analysis Grant BBS/E/ER/230002A and EI - Darwin Tree of Life Grant GP182.

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Decoding Living Systems